Andrew **Peterson**

Quantum Researcher · Security Engineer · Software Architect

California, United States of America

acpeters@ucsd.edu | andypeterson2 | i-am-andy-peterson

Education

University of California, San Diego *La Jolla, California*

Bachelor of Science in Computer Science *Sep. 2020 - Dec. 2024*

- Relevant Coursework: Advanced Data Structures, Design & Analysis of Algorithms, Operating Systems, Machine Learning, Computer Security, Networked Systems, Software Engineering, Discrete Mathematics

Experience

Qualcomm Institute *La Jolla, California*

Quantum Software Engineering Lead (Intern) *Jul. 2022 - Dec. 2024*

- Architected a quantum-encrypted video platform via photonic circuitry, achieving secure key distribution at 8 qubits with <4.2% QBER over 1.2-meter channels.
- Optimized Grover's algorithm in Qiskit for binary constraint satisfaction, achieving 4x speedup over classical brute-force on 16-variable problems with 38% circuit depth reduction.
- Built and benchmarked QML classification models via quantum neural networks, achieving 94.7% accuracy on Iris, outperforming classical baselines by 2.3 percentage points.
- Mentored a cross-functional team of 4 interns under Dr. Nikola Alic, overseeing software deliverables and commissioning a new undergraduate quantum computing lab.

Rochester Institute of Technology Esports *Remote (Rochester, New York)*

Full-Stack DevOps Engineer & Advisor *Sep. 2020 - Jul. 2022*

- Engineered and deployed MERN-stack web resources via Docker, serving 2,400 monthly active users across 12 varsity esports teams.
- Orchestrated cloud infrastructure on DigitalOcean Droplets, implementing Nginx reverse proxies and Docker Compose to ensure 99.9% uptime for high-traffic services.
- Built CI/CD pipelines and monitoring suites, reducing deployment time from 6 hours (manual) to 15 minutes and saving ~320 hours/year in release effort.

Mathnasium *Temecula, California*

Instructor & IT Systems Specialist *Mar. 2020 - Sep. 2021*

- Delivered advanced technical instruction in Calculus and Computer Science, tailoring complex concepts to diverse student learning levels.
- Directed the center's digital transformation during COVID-19, authoring technical documentation and maintaining IT infrastructure to enable seamless remote learning.

Extracurricular

Quantum Computing at UC San Diego *La Jolla, CA*

Co-Founder & President (2023 - 2024) & Technical Curriculum Lead *Feb. 2021 - Dec. 2024*

- Co-founded and scaled the first undergraduate quantum computing organization at UC San Diego, establishing a community for 100+ aspiring researchers and engineers.
- Developed a technical curriculum spanning 6 quarterly workshops, educating 200+ students with average attendance growing from 12 to 45.
- Standardized organizational workflows and digital infrastructure, enabling sustainable leadership transitions and cross-campus collaborations.
- Presented components of internship to organization members, including quantum-encrypted communication and quantum search algorithms.

ACM Cyber @ UCSD *La Jolla, California*

President & Cybersecurity Competition Member *Aug. 2020 - Jun. 2023*

- Trained teams for (and competed in) high-stakes national and international competitions including CCDC, CPTC, and Google CTF.
- Authored pedagogical training modules on penetration testing, network defense, and cryptography to elevate the club's competitive ranking.
- Orchestrated weekly technical workshops and "Deep Dive" sessions, fostering a culture of peer-to-peer mentorship in advanced security topics.

San Diego Capture the Flag (SDCTF) *Remote*

LEAD INFRASTRUCTURE ENGINEER & CHALLENGE DESIGNER *Dec. 2020 - May 2022*

- Engineered the backend for San Diego's largest annual cybersecurity competition, supporting 500+ global participants in its inaugural year.
- Provisioned cloud-native infrastructure using Terraform (IaC), managing a Postgres database and containerized challenge environments.
- Designed and deployed a Discord.JS-integrated submission bot to automate real-time flag validation and leaderboard updates.
- Developed high-quality Open Source Intelligence (OSINT) challenges, testing competitors' ability to perform investigative digital forensics.

Skills

Languages	Python (Expert), C/C++, Java, TypeScript/JavaScript, SQL, Bash, LaTeX, Matlab, Rust
Quantum Computing	Qiskit, Cirq, Quantum Neural Networks (QML), QAOA, Grover's & HHL Algorithms, BB84, Error Mitigation
Systems & DevOps	Docker, Kubernetes, Terraform, Linux Kernel, Distributed Systems, CI/CD, Nginx, Git, AWS, DigitalOcean
Cybersecurity	Reverse Engineering (Ghidra/IDA), Network Security (TCP/IP), Penetration Testing, OSINT, Applied Cryptography
Software Engineering	Data Structures & Algorithms (DSA), System Design, Design Patterns, Agile/Scrum, Unit Testing
Data Science & AI	PyTorch, Scikit-Learn, Pandas, Computer Vision, NLP, Ollama (LLMs), Statistical Modeling, Jupyter
Infrastructure	PostgreSQL, MongoDB, Redis, RESTful/GraphQL API Design, Microservices, Linux Server Administration
Experimental Lab	Free-space Photonics, Laser Systems, Signal Processing, Oscilloscopes, Atomic Force Microscopy (AFM)

Certifications

2025 **Azure Data Fundamentals**, Microsoft

2022 **IT Fundamentals+**, CompTIA

References

Available upon request.